Department of Computer Science and Engineering
National Institute of Technology Karnataka, Surathkal
242IS015

Mid-Semester Examination (February 2025)

Course Name: Quantum Cryptography
Semester: II
Duration: 1 Hour 30 Minutes

Course Code: CS826
Course: M.Tech. (CSE-IS)
Marks: 40

Note: Answer all the questions.

Q1. (a) Consider the complex number z=1+2i. **[08]**

- **(i)** Find the real component z of $\Re(z)$.
- **(ii)** Find the imaginary component z of $\Im(z)$.
- **(iii)** Write z in the polar form $re^{i\theta}$.
- **(iv)** Find the conjugate z*.

(b) A qubit is in the state **[04]**

$$\frac{1+i\sqrt{3}}{3}|0\rangle + \frac{2-i}{3}|1\rangle.$$

If you measure the qubit, what is the probability of getting
(a) $|0\rangle$?
(b) $|1\rangle$?

Q2. (a) A qubit is in the state

$$A\left(2e^{i\pi/6}|0\rangle - 3|1\rangle\right).$$

(i) Normalized the state (i.e., find A.). **[04]**

(ii) If you measure the qubit, what is the probability that you get $|0\rangle$? **[04]**

Q3. Consider a map U that transforms the Z-basis states as follows.

$$U|0\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{\sqrt{3}+i}{4}|1\rangle.$$

$$U|1\rangle = \frac{\sqrt{3}+i}{4}|0\rangle - \frac{\sqrt{3}+3i}{4}|1\rangle.$$

Say $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is a normalized quantum state, i.e., $|\alpha|^2 + |\beta|^2 = 1$.

(i) Calculate $U|\psi\rangle$. **[05]**

(ii) From your answer to (i), is U a valid gate? Explain your reasoning. **[05]**

Scanned with OKEN Scanner

Q4. (a) The Hadamard gate turns $|0\rangle$ into $|+\rangle$, and $|1\rangle$ into $|-\rangle$.

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle,$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle.$$

From the definition of Hadamard gate, prove that H|+> = |0>, H|-> = |1> and H|i> = |-i>. **[02+02+02]**

(b) Consider each logical classic gate with inputs A and B, outputs C and D, and the truth table below. Is each gate a valid quantum gate? Why? **[02+02]**

(a)

A	B
0	0
i	1

(b)

A	B
0	1
1	1

C D

$H|i\rangle = |-i\rangle$

$|00\rangle + |11\rangle$

$|01\rangle + |11\rangle$

Scanned with OKEN Scanner

Department of Computer Science and Engineering
National Institute of Technology Karnataka, Surathkal

End-Semester Examination (May 2025)

Course Name: Quantum Cryptography
Semester: II
Duration: 03 Hours

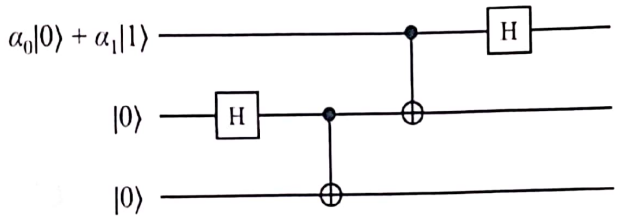
Course Code: CS826
Course: M.Tech. (CSE-IS)
Marks: 80

Note: Answer all the questions.

Q1. (a) Explain Shor's Algorithm and describe through a suitable example how it can pose a threat to RSA. **[6]**

(b) A SWAP gate takes two inputs x_1 and x_2 and outputs x_2 and x_1; i.e., it swaps the values of two registers. Show how to build a SWAP gate using only CNOT gates. **[6]**

Q2. Consider the following quantum circuit:



An equivalent description of the circuit (calling the registers x_1, x_2, x_3) is:

1. Initialize x_1 to $\alpha_0 |0\rangle + \alpha_1 |1\rangle$
2. Initialize x_2 to $|0\rangle$
3. Initialize x_3 to $|0\rangle$
4. Hadamard(x_2)
5. CNOT(x_2, x_3)
6. CNOT(x_1, x_2)
7. Hadamard(x_1)

a) Determine with proof the state of the three qubits at the end of the circuit's operation. **[05]**

b) If we then measure the three qubits, give the outcomes and their probabilities that arise. **[05]**

Q3. Calculate the following inner products. **[03+03+03]**

a) $\langle 10|11\rangle$.

b) $\langle + -|01\rangle$.

c) $\langle 1+0|1-0\rangle$.

Scanned with OKEN Scanner

Q4. (a) If you measure the left qubit, what outcomes can you get, what are the corresponding probabilities of those outcomes, and what does the state collapse to for each outcome? Is this state a product state, partially entangled state, or maximally entangled state? **[03+03+03]**

$$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle).$$

(b) Alice and Bob share an EPR-pair,

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Suppose they each measure their qubit with an X-observable (which corresponds to a particular projective measurement with possible outcomes +1, −1). Find the following. **[10]**

(a) P_{00}.

(b) P_{01}.

(c) P_{10}.

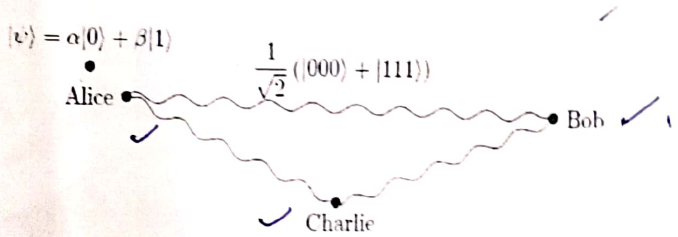
(d) P_{11}.

(e) $E(A',B') = P00 - P01 - P10 + P11$.

Q5. Alice wants to teleport a qubit in an unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to Charlie, and Bob is helping her. They share three entangled qubits in the GHZ state:

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle).$$

The left qubit is Alice's, the middle qubit is Bob's, and the right qubit is Charlie's.



Altogether, the initial state of the system is

$$|\psi\rangle|GHZ\rangle = (\alpha|0\rangle + \beta|1\rangle)\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

$$= \frac{1}{\sqrt{2}}(\alpha|0000\rangle + \alpha|0111\rangle + \beta|1000\rangle + \beta|1111\rangle$$

Scanned with OKEN Scanner

So, the left two qubits are Alice's, the second-to-right qubit is Bob's, and the right qubit is Charlie's.

(a) Show that if Alice applies CNOT to her qubits (so the far left qubit is the control and the second-to-left qubit is the target) and then the Hadamard gate to her left qubit, the state of the system becomes **[5]**

$$\frac{1}{2}\big[|00\rangle(\alpha|00\rangle+\beta|11\rangle)+|01\rangle(\beta|00\rangle+\alpha|11\rangle)$$
$$+|10\rangle(\alpha|00\rangle-\beta|11\rangle)+|11\rangle(-\beta|00\rangle+\alpha|11\rangle)\big].$$

(b) Next, Alice measures her two qubits and makes the results known. What values can she get, with what probabilities, and what does the state collapse to in each case? **[5]**

Q6. (a) Explain the Crystal-Kyber public encryption algorithm. **[6]**

(b) Show that the following gates are reversible or irreversible. **[4]**

(i) OR. (ii) XOR. (iii) NAND. (iv) NOR

Q7. (a) Let's assume:

Alice's key:

A = 1 0 1 1 0 1 1 0

Bob's key:

B = 1 0 1 1 1 1 1 0

The above Alice and Bob raw keys contain discrepancies. Demonstrate how the Cascade protocol can be used to detect and correct errors. **[5]**

(b) Explain the BB84 Quantum Key Distribution protocol with the help of a suitable example. **[5]**

Scanned with OKEN Scanner

252PS032

Department of Computer Science and Engineering
National Institute of Technology Karnataka, Surathkal

Mid-Semester Examination (February 2026)

Course Name: Quantum Cryptography
Semester: II
Duration: 1 Hour 30 Minutes
Note: Answer all the questions.

Course Code: CS826
Course: M.Tech. (CSE-IS)
Marks: 25

Q1. (a) A qubit is in the following state: [2]

$$\begin{pmatrix} \sqrt{3}/2 \\ 1/2 \end{pmatrix}$$

If you measure this qubit in the Z-basis $\{|0\rangle, |1\rangle\}$, what states can you get and with what probabilities?

(b) Consider

$$|a\rangle = \frac{3+i\sqrt{3}}{4}|0\rangle + \frac{1}{2}|1\rangle,$$
$$|b\rangle = \frac{1}{4}|0\rangle + \frac{\sqrt{15}}{4}|1\rangle.$$

(i) Find $\langle a|b\rangle$. [1]
(ii) Find $\langle b|a\rangle$. [1]
(iii) What is the relationship between your answers to parts (a) and (b)? [1]

(c) Consider a qubit in the following state [4]

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle.$$

Consider measuring this qubit in the Y-basis $\{|i\rangle, |-i\rangle\}$ and the orthonormal basis $\{|a\rangle, |b\rangle\}$, where

$$|a\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle,$$
$$|b\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

(i) Calculate $\langle i|\psi\rangle$.
(ii) Calculate $\langle -i|\psi\rangle$.
(iii) If you measure the qubit in the Y-basis, what states can you get and with what probabilities?
(iv) Calculate $\langle a|\psi\rangle$.
(v) Calculate $\langle b|\psi\rangle$.
(vi) If you measure the qubit in the $\{|a\rangle, |b\rangle\}$ basis, what states can you get and with what probabilities?

Q2. (a) The following two states are opposite points on the Bloch sphere:

$$|a\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle,$$
$$|b\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle.$$

So, we can measure relative to them. Now consider a qubit in the state

$$|\varphi\rangle = \frac{1}{2}|0\rangle - \frac{\sqrt{3}}{2}|1\rangle.$$

(i) Write the qubit's state in terms of $|a\rangle$ and $|b\rangle$. [2]

(ii) If you measure the qubit in the basis $\{|a\rangle, |b\rangle\}$, what states can you get and with what probabilities? [2]

(b) Consider the following two states

$$|a\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle,$$
$$|b\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle.$$

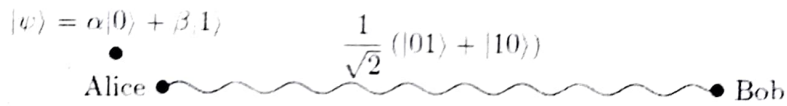
Prove these are opposite points of the Bloch sphere by finding their points in spherical coordinates (θ_a,φ_a) and (θ_b,φ_b). Verify that $\theta_b = \pi - \theta_a$ and $\varphi_b = \varphi_a + \pi$, which means they lie on opposite points of the Bloch sphere. [2]

Q3. Consider the following state of two qubits:

$$\frac{\sqrt{3}}{2\sqrt{2}}|00\rangle + \frac{1}{2\sqrt{2}}|01\rangle + \frac{1}{2\sqrt{2}}|10\rangle + \frac{\sqrt{3}}{2\sqrt{2}}|11\rangle$$

If you measure the left qubit, what outcomes can you get, what are the corresponding probabilities of those outcomes, and what does the state collapse to for each outcome? Is this state a product state, partially entangled state, or maximally entangled state? [1+1+1+1]

Q4 Alice wants to teleport a qubit in an unknown state $|\psi\rangle = \alpha|0\rangle+\beta|1\rangle$ to Bob. Instead of sharing two entangled qubits in the $|\Phi+\rangle$ state, they share two entangled qubits in the $|\Psi+\rangle$ state:



Altogether, the initial state of the system is

$$|\psi\rangle|\Psi^+\rangle = (\alpha|0\rangle + \beta|1\rangle)\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$
$$= \frac{1}{\sqrt{2}}(\alpha|001\rangle + \alpha|010\rangle + \beta|101\rangle + \beta|110\rangle).$$

So, the left two qubits are Alice's, and the right qubit is Bob's.

(i) Show that if Alice applies CNOT to her two qubits, followed by H to her left qubit, the state of the system becomes [2]

$$\frac{1}{2}\big[|00\rangle(\beta|0\rangle + \alpha|1\rangle) + |01\rangle(\alpha|0\rangle + \beta|1\rangle)$$
$$+ |10\rangle(-\beta|0\rangle + \alpha|1\rangle) + |11\rangle(\alpha|0\rangle - \beta|1\rangle)\big].$$

(ii) Next, Alice measures both of her qubits. What values can she get, with what probabilities, and what does the state collapse to in each case? [2]

(ii) Finally, Alice tells Bob the results of her measurement. For each possible result, what should Bob do to his qubit so that it is $\alpha|0\rangle+\beta|1\rangle$, the state that Alice wanted to teleport to him? [2]